How Vectors Work in Quality AI

A complete visual walkthrough — from a sentence to a number, and from a number to an answer

384-Dimensional Embeddings | all-MiniLM-L6-v2 AI Model | Cosine Angle Mathematics | Semantic RAG Search

What is a Vector? The Core Idea

💡 The Simplest Explanation

A **Vector** is just a **list of numbers** that represents the *meaning* of a sentence. Computers cannot read words — but they can compare numbers perfectly. So the AI model converts every quality problem sentence into 384 numbers. Sentences with similar engineering meaning will produce number-lists that **point in the same mathematical direction**.

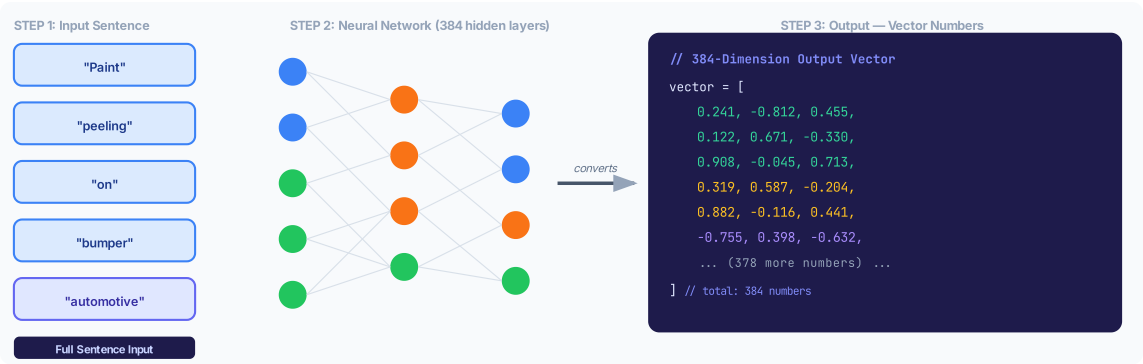

Figure 1 — Every sentence is fed through the all-MiniLM-L6-v2 neural network and comes out as a list of 384 numbers (a Vector). These numbers capture the full meaning of the text.

☀️ Why This Matters

"Paint peeling on automotive bumper" → `[0.241, -0.812, 0.455, ...]`
"Coating detachment on vehicle panels" → `[0.235, -0.798, 0.461, ...]`

Notice how **completely different words** produce **almost identical number-lists**? That is the power of vector embeddings — they capture engineering concepts, not just spelling.

The Vector Space — How Similarity Works Visually

After converting every sentence to a vector, the AI places each one as an **arrow** in mathematical space. Arrows pointing in the **same direction = similar meaning**. Arrows pointing sideways = unrelated.

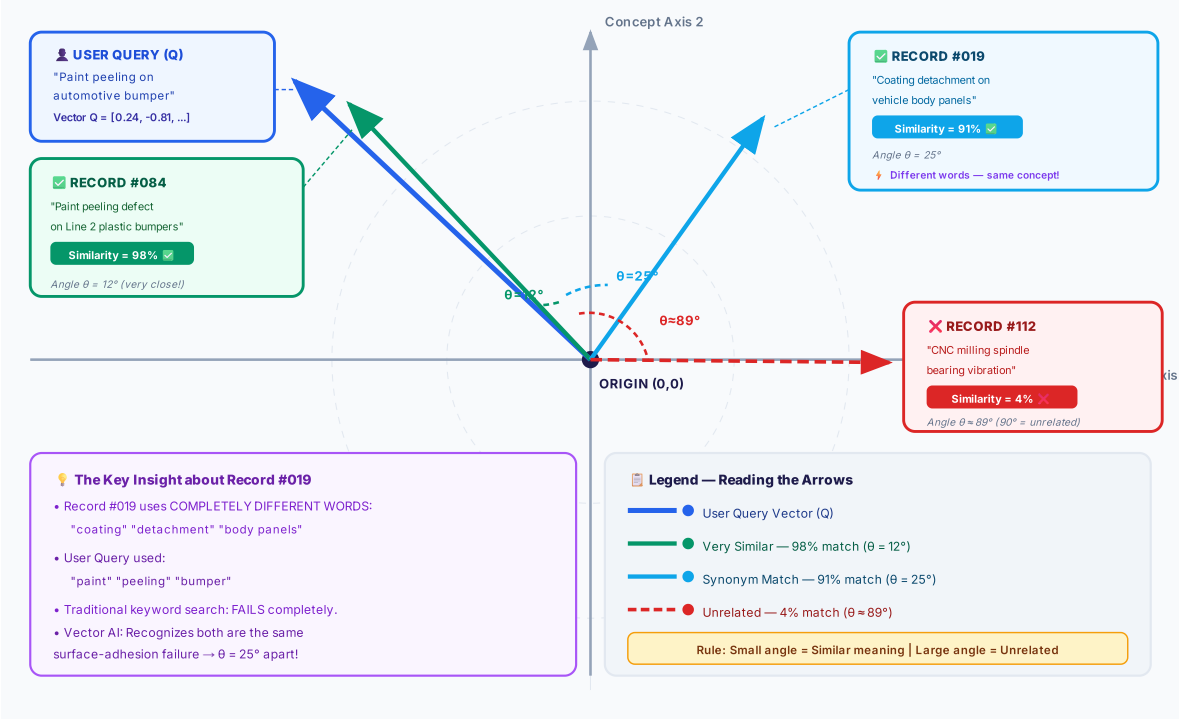

Figure 2 — 2D vector space diagram. Each sentence is an arrow from the Origin. Arrows pointing close together = similar meaning. All label cards are fully separated with no overlap.

The Cosine Formula — Step by Step

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// The Formula — measures the angle between two vectors: Similarity(Q, V) = (Q · V) / (||Q|| × ||V||) // Where: // Q · V = Dot Product (multiply matching numbers, then add all) // ||Q|| = Magnitude of Query (square-root of sum of squares) // ||V|| = Magnitude of the Past Project vector // // Result range: 1.0 = identical meaning | 0.0 = completely unrelated
```

Worked Example with 3 Numbers (Same formula, simplified from 384)

Vector	Sentence	Numbers (simplified 3D)
Q (User Query)	"Paint peeling on automotive bumper"	`[1, 2, 3]`
V_1 (Project #084)	"Paint peeling defect on plastic bumpers"	`[2, 4, 6]` — same direction, scaled 2×
V_2 (Project #112)	"CNC milling spindle bearing vibration"	`[5, -2, 0]` — different direction entirely

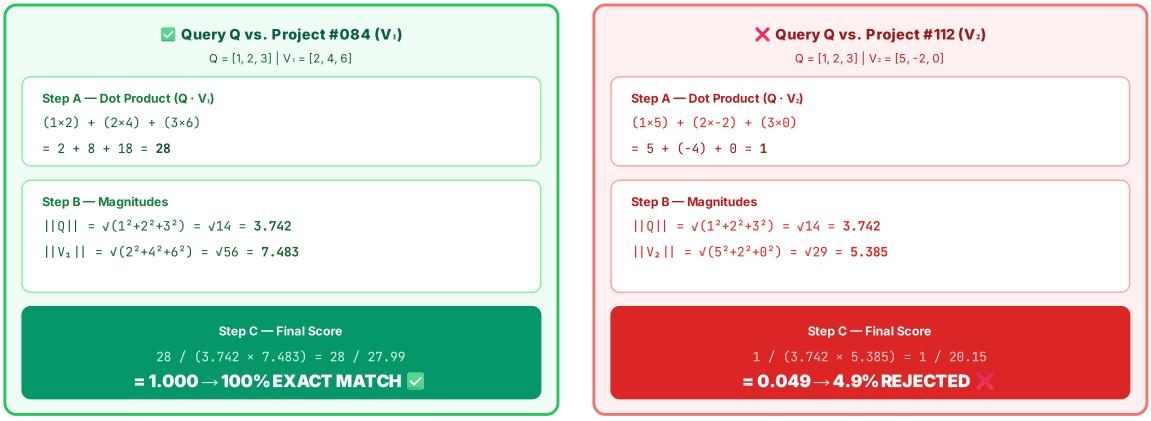

Figure 3 — Side-by-side calculation showing why Project #084 scores 100% (vectors point same direction) and Project #112 scores only 4.9% (vectors point sideways, nearly 90° apart)

Complete Data Flow — From Question to Answer

1 👤 **Engineer types a question in the AI chat widget**
"How was paint peeling on automotive bumpers solved before?"
`POST /api/rag/chat → { query: "How was paint peeling..." }`

2 🔒 **JWT Auth + Organization Lock**
User identity verified. ALL database queries are locked to `org_id = 5` so no data leaks across tenants.

3 🧠 **Sentence → 384-Number Vector Embedding**
The `all-MiniLM-L6-v2` model converts the full sentence into a 384-dimensional vector in ~2ms.
`Query Vector Q = [0.241, -0.812, 0.455, 0.122, 0.671, ... (384 total)]`

4 🗄️ **Fetch All Past Project Embeddings from PostgreSQL**
Queries `knowledge_repository` table. Every past project already has its pre-stored 384-number vector saved during data ingestion.
`SELECT id, title, problem_summary, root_cause, embedding FROM knowledge_repository WHERE org_id = 5`

5 📐 **Cosine Similarity Computed for Every Past Project**
For each stored project vector V, the system computes: `Similarity = (Q·V) / (||Q||×||V||)`. All results sorted 1.0→0.0 (highest relevance first).

6 ✨ **Top 4 Matches Formatted into a Verified Answer**
Root cause, countermeasure, KPI improvement % and cost savings ₹ from the top matches are assembled into a clean markdown response with clickable project storyboard links.

Ranked Results Visualization

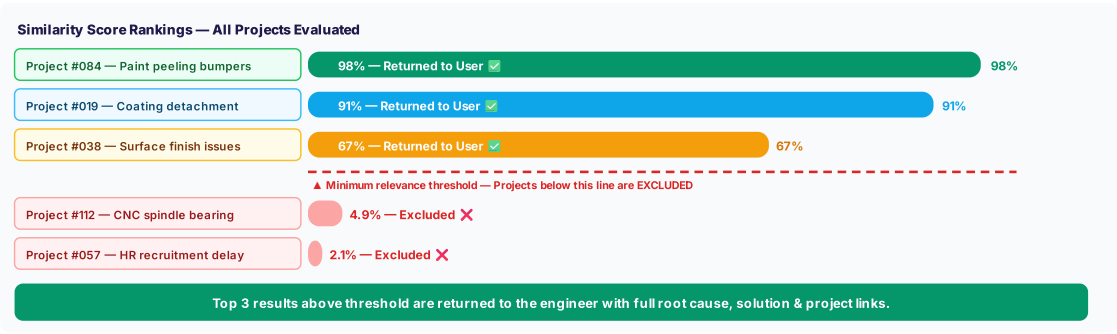

Figure 4 — Every past project receives a mathematical score. Only the top relevant matches (above threshold) are shown to the engineer.